

Evaluation of the Video Ocular Counter-Roll (vOCR) as a New Clinical Test of Otolith Function in Peripheral Vestibulopathy

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IMPORTANCE Video-oculography (VOG) goggles have been integrated into the assessment of semicircular canal function in patients with vestibular disorders. However, a similar bedside VOG method for testing otolith function is lacking.

OBJECTIVE To evaluate the use of VOG-based measurement of ocular counter-roll (vOCR) as a clinical test of otolith function.

DESIGN, SETTING, AND PARTICIPANTS A case-control study was conducted to compare vOCR measurement among patients at various stages of unilateral loss of vestibular function with healthy controls. The receiver operating characteristic curve method was used to determine the diagnostic accuracy of the vOCR test in detecting loss of otolith function. Participants were recruited at a tertiary center including the Johns Hopkins outpatient clinic and Johns Hopkins Hospital, Baltimore, Maryland. Participants included 56 individuals with acute (≤ 4 weeks after surgery), subacute (4 weeks–6 months after surgery), and chronic (> 6 months after surgery) unilateral vestibular loss as well as healthy controls. A simple bedside maneuver with en bloc, 30° lateral tilt of the head and trunk was used for vOCR measurement. The study was conducted from February 2, 2017, to March 10, 2019.

INTERVENTION In each participant vOCR was measured during static tilts of the head and trunk en bloc.

MAIN OUTCOMES AND MEASURES The vOCR measurements and diagnostic accuracy of vOCR in detecting patients with loss of vestibular function from healthy controls.

RESULTS Of the 56 participants, 28 (50.0%) were men; mean (SD) age was 53.5 (11.4) years. The mean (SD) time of acute unilateral vestibular loss was 9 (7) days (range, 2–17 days) in the acute group, 61 (39) days (range, 28–172 days) in the subacute group, and 985 (1066) days (range 185–4200 days) in the chronic group. The vOCR test showed reduction on the side of vestibular loss, and the deficit was greater in patients with acute and subacute vestibular loss than in patients with chronic loss and healthy controls (acute vs chronic: -1.81° ; 95% CI, -3.45° to -0.17° ; acute vs control: -3.18° ; 95% CI, -4.83° to -1.54° ; subacute vs chronic: -0.63° ; 95% CI, -2.28° to 1.01° ; subacute vs control: -2.01° ; 95% CI, -3.65° to -0.36° ; acute vs subacute: -1.17° ; 95% CI, -2.88° to 0.52° ; and chronic vs control: -1.37° ; 95% CI, -2.96° to 0.21°). The asymmetry in vOCR between the side of vestibular loss and healthy side was significantly higher in patients with acute vs chronic loss (0.28; 95% CI, 0.06–0.51). Overall, the performance of the vOCR test in discriminating between patients with vestibular loss and healthy controls was 0.83 (area under the receiver operating characteristic curve). The best vOCR threshold to detect vestibular loss at the 30° tilt was 4.5° , with a sensitivity of 80% (95% CI, 0.62%–0.88%) and specificity of 82% (95% CI, 0.57%–1.00%).

CONCLUSIONS AND RELEVANCE The findings of this case-control study suggest that the vOCR test can be performed with a simple bedside maneuver and may be used to detect or track loss of otolith function.

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Video-oculography (VOG) goggles have been integrated into the bedside assessment of patients with vestibular disorders. The emphasis has been mostly on the function of semicircular canals using the video head impulse test (vHIT), but a similar VOG method for the clinical assessment of otolith function is lacking.^{1,2} The otolith organs detect inertial forces on the head and provide vestibular inputs essential for a wide range of neurophysiologic functions, such as the vestibulo-ocular reflex (VOR) during head tilt or head translation, sensing head motion and position, postural control, and spatial orientation.³⁻⁹ Because otolith pathways are commonly affected in vestibular disorders, an easy-to-use, bedside VOG test of otolith function would be valuable in the evaluation of these patients.¹⁰⁻¹²

When the head is tilted laterally in the roll plane (ear to shoulder), the resulting response is a torsional rotation of the eyes in the opposite direction of the head tilt, known as the ocular counter-roll (OCR).¹³⁻¹⁵ During the movement of the head, the dynamic component of the OCR consists of a torsional nystagmus with a slow phase in the opposite direction of the head tilt, which is driven by the activity from the semicircular canals and otolith organs.^{16,17} During a sustained head tilt, however, the static OCR maintains the ocular torsion away from the side of the head tilt, which is primarily generated by the otolith inputs from the utricles.^{3,18-20}

In this study, we sought to examine whether VOG-based OCR (vOCR) measurement can be used to detect loss of otolith function at the bedside. For vOCR recording, we used a video method that can track torsional eye position accurately in real-time using an iris-recognition algorithm.¹⁵ In a previous study, the vOCR measurement using this video method was able to identify patients with chronic vestibular loss, and the accuracy was comparable to the vestibular-evoked myogenic potentials (VEMP), a widely used laboratory test of otolith function.^{15,21} The vOCR was symmetrically reduced between the side of vestibular loss and the healthy side in patients with chronic unilateral vestibular loss. In this study, to evaluate vOCR as a clinical test, we applied a simple bedside maneuver during which the head and trunk were tilted laterally en bloc by the examiner. We evaluated the accuracy of this bedside method in detecting abnormal vestibular function at various stages following vestibular loss.

Methods

We recruited patients with unilateral loss of vestibular function from a resection of vestibular schwannoma (Table). Patients were enrolled consecutively from the Johns Hopkins outpatient clinic or during their hospital stay at Johns Hopkins Hospital, Baltimore, Maryland, after their vestibular schwannoma was resected. All participants gave written informed consent and the study was approved by the Johns Hopkins Medicine Institutional Review Board. Participants did not receive financial compensation. The study was conducted from February 2, 2017, to March 10, 2019. This study followed the Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for case-control

Key Points

Question Can a video oculography-based ocular counter-roll measurement be used as a clinical test of otolith function?

Findings In this case-control study of 56 individuals, video ocular counter-roll measurement with a simple bedside head tilt maneuver could detect patients with loss of vestibular function and healthy controls. The technique had a diagnostic accuracy of 83%.

Meaning The video ocular counter-roll (vOCR) test can be performed with a simple bedside maneuver to detect or track loss of otolith function.

studies. Patient recruitment was based only on unilateral loss of vestibular function regardless of whether they had any vestibular symptom or sign of vestibular recovery. Individuals with any other neurologic abnormality (eg, ischemic lesions noted on brain magnetic resonance imaging) or history of vestibular or hearing loss on the nonlesion side were not included. Forty-five patients were initially recruited but 4 were later excluded due to difficulty completing the maneuvers after the surgery or poor VOG data quality. Based on the time from resection of vestibular schwannoma, patients were divided into 3 subgroups: within 4 weeks after surgery (acute, $n = 13$), between 4 weeks and 6 months after surgery (subacute, $n = 13$) and more than 6 months after surgery (chronic, $n = 15$). All patients had loss of hearing and signs of vestibular loss on the side of the surgery: mean (SD) horizontal vHIT gain for acute on the lesion side, 0.28 (0.19) vs 0.80 (0.13) on the nonlesion side; subacute on the lesion side, 0.33 (0.11) vs 0.69 (0.18) on the nonlesion side; and chronic on the lesion side, 0.44 (0.16) vs 0.83 (0.30) on the nonlesion side. A healthy control group ($n = 15$) was also recruited with no known neurologic, auditory, or vestibular disorders and with normal vestibular evaluation (mean [SD] horizontal vHIT gain on both sides, 0.91 [0.10]). All 4 groups of participants were age-matched (Table). A total of 56 individuals were included in the analysis.

vOCR Measurement and Analysis

A bedside tilt maneuver was applied for vOCR measurement, during which the head and trunk were tilted en bloc 30° while the participant was sitting upright (Figure 1). We used this maneuver because, when the head and trunk are tilted together, the static OCR is primarily driven by the otolith inputs, whereas if the head is tilted on the trunk, a change of inputs from the neck proprioceptors in addition to the vestibular inputs may contribute to the static OCR response.²²⁻²⁵ The degree of tilt was measured by a linear accelerometer mounted on the VOG goggles and the operator received numeric feedback on the head roll and pitch angles in real time so that a roll angle of 30° and a pitch angle of 0° could be maintained during the tilt maneuver. We chose 30° lateral tilt because it is large enough to produce a measurable OCR and yet within the comfortable range of positions to maintain during the tilt maneuver.

Participants sat upright on a chair, fixing on a visual target (2.5 cm in diameter), 135 cm away at eye level. The torsional position of both eyes and the position of the head were

Table. Patient Groups Based on the Time From Unilateral Loss of Vestibular Function

Variable	Time from lesion development, mean postoperative d (range)	Sex, No. (%)		Age, mean (SD), y	vOCR, mean (SD), degrees [°]	
		Women	Men		Lesion side	Nonlesion side
Acute	9 (2-17)	6 (46.2)	7 (53.8)	54.1 (10.8)	2.19 (1.79)	3.80 (1.93)
Subacute	61 (28-172)	7 (53.8)	6 (46.2)	52.8 (13.2)	3.36 (1.41)	3.89 (1.36)
Chronic	985 (185-4200)	8 (53.3)	7 (46.7)	54.5 (11.3)	4.00 (1.91)	4.00 (1.64)
Control	NA	7 (46.7)	8 (53.3)	52.5 (11.4)	5.37 (1.35)	5.43 (1.55)

Abbreviations: NA, not applicable; vOCR, video-oculography goggle-based measurement of ocular counter-roll.

recorded simultaneously using the VOG goggles. The recording of ocular torsion and head positions began with the head and trunk in the upright position for 30 seconds, followed by 3 right and 3 left lateral tilts in random order, each lasting 30 seconds, separated by 30-second periods with the body back in the upright position (Figure 1). There was also a final 30-second recording during which the participant was returned to the upright position from the last tilt position. A soft collar was used to restrict the movement of the neck on the trunk. The soft collar was equipped with a protractor to ensure that the head and trunk were aligned, and both were tilted at 30° during the maneuver. In addition to the fixation target straight ahead, 2 lateral targets were used so that the straight-ahead fixation could be maintained during the tilt by instructing the participants to fix on the lateral target. All 3 targets were in line with the lateral targets placed at equal distances from the central target. The distance between the lateral and central targets was calculated as the difference between the sitting surface (participant's hip level) to the lateral canthus of the eyes divided by 2 (ie, multiplied by the sine of 30°), hence taking into account the height of the trunk for maintaining straight-ahead fixation during 30° tilts. Thus, the taller the trunk, the farther the lateral targets had to be displaced to the sides.

We used a videonystagmoscopy system (the RealEyes xDVR goggles; Micromedical Technologies Inc) for vOCR measurement. To measure the tilt angle simultaneously, an accelerometer (MPU-92/65 sensor; InvenSense) was mounted onto the goggles and connected through an Arduino board (Teensy 3.2) to the input port of the cameras. We used custom software previously developed to measure ocular torsion.¹⁵ To determine the angle of ocular torsion with this method, a template-matching method is implemented to compare the iris pattern at any point in time during head tilt with a reference image obtained at the beginning of the recording with the head in the upright position (Figure 2). Further technical details of this method have been published previously.¹⁵

Data Analysis

Ocular torsion and tilt position data were analyzed in Matlab, version 9.7, 9.8, and 9.9 (Mathworks Inc). The vOCR value was measured as the difference between the average ocular torsion of both eyes during the static tilt from the preceding reference value in the upright position. The vOCR was measured with lateral en bloc tilts to each side (ie, right and left tilts). The final vOCR value was the mean of 3 measurements in each tilt direction. We only used the ocular torsion values during the static tilt position to avoid canal stimulation dur-

ing vOCR measurement. Because the static tilt was maintained manually and there could be a few degrees of difference in the actual tilt position vs the desired position of 30° during the test, the vOCR value was corrected based on the actual amount of static tilt so that the values could be compared across participants:

$$\text{Corrected vOCR} = (\text{measured vOCR} \times 30) / (\text{measured tilt}).$$

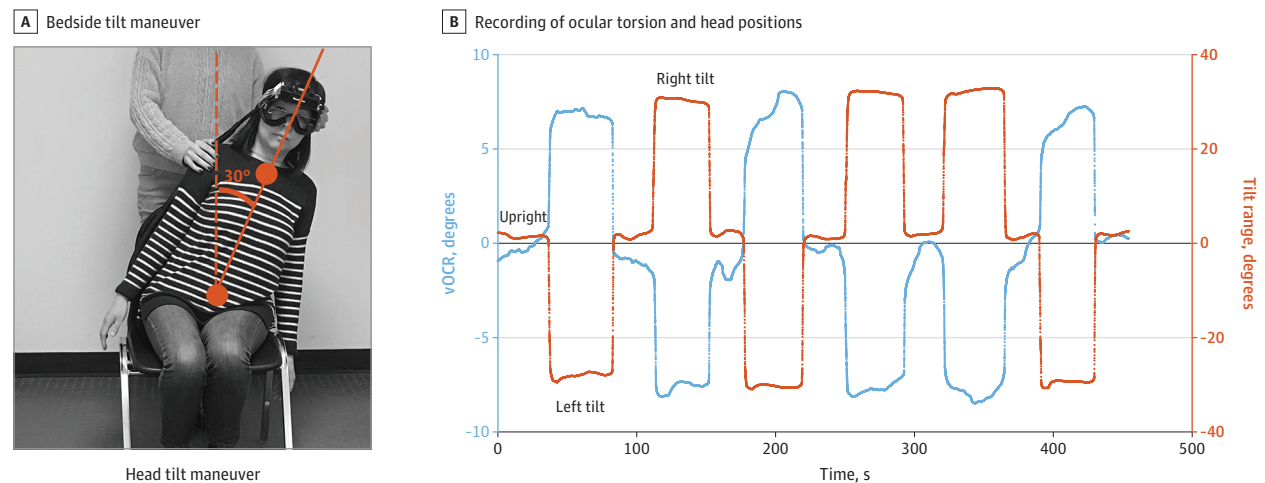
To account for the difference between the vOCR values on the lesion side (ie, the side of vestibular loss or LS) and the nonlesion side (NLS), the vOCR asymmetry ratio was calculated as:

$$\text{vOCR asymmetry ratio} = (\text{NLS} - \text{LS}) / (\text{abs}[\text{LS}] + \text{abs}[\text{NLS}]).$$

Statistical Analysis

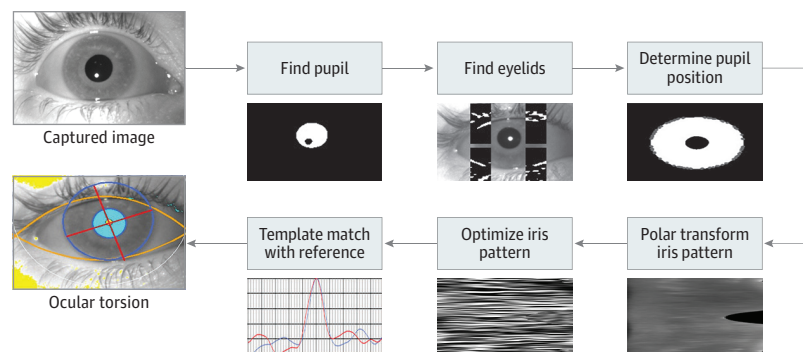
Statistical analyses were done in Matlab, versions 9.7, 9.8, and 9.9 and R, version 4.0.4 (R Foundation for Statistical Computing) statistical software using the effect size and 95% CI. To compare the mean vOCR during head tilt and account for multiple comparisons, we conducted 2-way repeated measures analysis of variance on the influence of 2 independent variables: participant groups, including healthy controls and patients with acute, subacute, and chronic vestibular loss; and the side tested, including lesion and nonlesion sides in patients and 2 randomly assigned sides among controls (Table). The asymmetry ratios among the participant groups were compared using 1-way analysis of variance. Variables in the analysis of variance were compared using the η^2 test, which quantifies the variation in the results by a given variable. Generally, η^2 values used to categorize the variations are 0.01 (small), 0.06 (medium), and 0.16 or greater (large).

The accuracy of the vOCR test in discriminating between vestibular loss and normal function was evaluated using receiver operating characteristic (ROC) analysis by including all participants (41 patients and 15 healthy controls). In this method, the sensitivity (true-positive rate) is plotted against 1-specificity (false-positive rate) for different thresholds of a test parameter. Thus, each point on the ROC curve corresponds to the true- and false-positive rates at a particular decision threshold. In the ROC analysis, the area under the curve is a measure of the test's ability to distinguish between patients with vestibular loss and healthy controls, and the best threshold can be determined as the point on the ROC curve representing the best sensitivity and specificity for the test. For ROC analysis, vOCR was the index test and the target condition was loss of vestibular function. The clinical reference was

Figure 1. Tilt Maneuver as a Passive 30° Tilt of the Head and Trunk While the Participant Is Sitting on a Chair

The recording started in the upright position followed by 3 right and 3 left tilts in random order separated by a return to the upright position during which video-oculography measurement of ocular counter-roll (vOCR) and head tilt angle were measured simultaneously. Note the different scales on the ordinate

for the tilt angle and vOCR. The right panel shows an example of the vOCR recording (blue trace/left axis) during the en bloc tilt maneuver (tilt angle shown by the orange trace/right axis). The positive value on the graph indicates a rightward direction and the negative value indicates a leftward direction.

Figure 2. General Structure of the Video-Oculography (VOG) Method Used to Detect Ocular Torsion and Measure Ocular Counter-Roll (OCR)

This torsion method is based on template matching of the entire iris and tracks the eyelids to account for partial occlusion of the iris and pupil.¹⁵ A polar transformation is applied to the iris pattern and the image is optimized to enhance the iris features and mask the parts covered by the eyelids. For OCR

measurement, the iris pattern is set as the baseline reference in the upright position and a template-matching method is implemented to compare the iris pattern at any given point in time during head tilt with the baseline reference in the upright position.

the known state of vestibular function in patients and healthy controls.

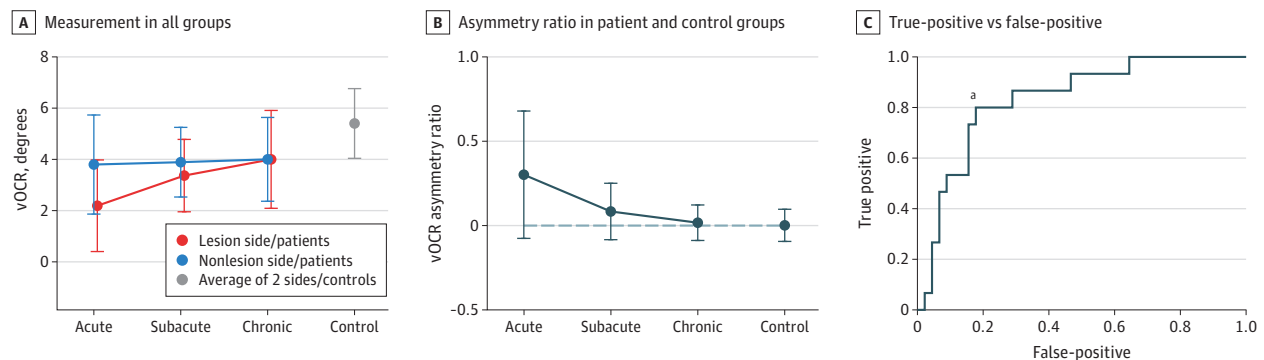
Results

Comparisons of vOCR Among Participant Groups

The population included 28 men (50.0%) and 28 women (50.0%); mean (SD) age 53.5 (11.4) years. Other demographic information and vOCR values for all patient and control groups are provided in the Table. The groups that represented the time from loss of vestibular function were significantly associated with the vOCR values (2-way repeated measures analysis of variance; $\eta^2 = 0.18$; 95% CI, 0.006-0.35) (Figure 3A). Post hoc comparisons showed that the mean (SD) vOCR on the lesion

side in the acute loss group was lower than in the chronic loss and control groups (acute, 2.19° [1.79°]; subacute, 3.36° [1.41°]; chronic, 4.00° [1.91°]; and control, 5.37° [1.35°]). In between-group comparisons, decreases were noted with acute vs chronic: -1.81°; 95% CI, -3.45° to -0.17°; and acute vs control: -3.18°; 95% CI, -4.83° to -1.54°; however, the difference from the subacute patient group was not significant (acute vs subacute: -1.17°; 95% CI, -2.88° to 0.52°) (Figure 3A). Also, the vOCR on the lesion side in the subacute group was lower than the control group (subacute vs control: -2.01°; 95% CI, -3.65° to -0.36°), but it was not significantly different from the chronic group (subacute vs chronic: -0.63°; 95% CI, -2.28° to 1.01°) (Figure 3A). The vOCR on the lesion side in the chronic group was not lower than the control group (chronic vs control: -1.37°; 95% CI, -2.96° to 0.21°). The vOCR on the nonlesion side was

Figure 3. Video-Oculography Measurement of Ocular Counter-Roll (vOCR) Results



A, Mean vOCR measurements in all participant groups. B, Mean vOCR asymmetry ratios in patient and control groups. Error bars represent SD. C, Receiver operating characteristic curve from the average vOCR measurements from both sides in all patients and healthy controls. The area

under the curve was 0.83.

^a The best vOCR threshold to detect vestibular loss was 4.5° with a sensitivity of 80% and specificity of 82%.

not significantly different by group (mean [SD], acute, 3.80° [1.93°]; subacute, 3.89° [1.36°]; chronic, 4.00° [1.64°]; and control, 5.43° [1.55°]) and between groups (acute vs chronic: -0.20°; 95% CI, -1.84° to 1.44°; acute vs control: -1.63°; 95% CI, -3.27° to 0.01°; subacute vs control: -1.54°; 95% CI, -3.18° to 0.10°; subacute vs chronic: -0.11°; 95% CI, -1.75° to 1.53°; and chronic vs control: -1.43°; 95% CI, -3.01° to 0.15°).

Comparisons of vOCR on the Lesion and Nonlesion Sides

There was a significant association of the measurement side and the vOCR values ($\eta^2 = 0.34$; 95% CI, 0.11-0.52). There was also a significant interaction between the side of vOCR measurement and participant groups ($\eta^2 = 0.24$; 95% CI, 0.04-0.41). Similarly, the participant group had a significant association with the asymmetry ratio as the measure of difference in vOCR between the lesion and nonlesion sides (mean [SD] acute, 0.30 [0.38]; subacute, 0.08 [0.17]; and chronic, 0.02 [0.11]; $\eta^2 = 0.22$; 95% CI, 0.004-0.46) (Figure 3B). Post hoc comparisons showed that the lesion-side vOCR was lower than the nonlesion-side vOCR in the acute patient group (-1.61°; 95% CI, -2.32° to -0.89°) (Figure 3A). No significant difference was found between the vOCR value on either side in the subacute, chronic, or control groups (subacute: -0.52°; 95% CI, -1.24° to 0.19°; chronic -0.001°; 95% CI, -0.67° to 0.66°; and control: -0.05°; 95% CI, -0.72° to 0.61°). The vOCR asymmetry in the acute group was higher than in the chronic group (0.28; 95% CI, 0.06-0.51), but there was no significant difference between the acute and subacute groups (0.22; 95% CI, -0.01 to 0.45) or between the subacute and chronic groups (0.07; 95% CI, -0.16 to 0.29).

vOCR Diagnostic Accuracy

We used ROC analysis to evaluate the performance of the vOCR test (index test) using the mean vOCR measurements from both sides in patients and healthy controls (Figure 3C). The performance of the test in discriminating between patients with vestibular loss and healthy controls, determined by the area under the curve, was 0.83. The best vOCR threshold to detect

vestibular loss was 4.5°, with a sensitivity of 80% (95% CI, 0.62%-0.88%) and specificity of 82% (95% CI, 0.57%-1.00%) (Figure 3C). At this threshold, the true-positive value was 33; true-negative, 12; false-positive, 3; and false-negative, 8.

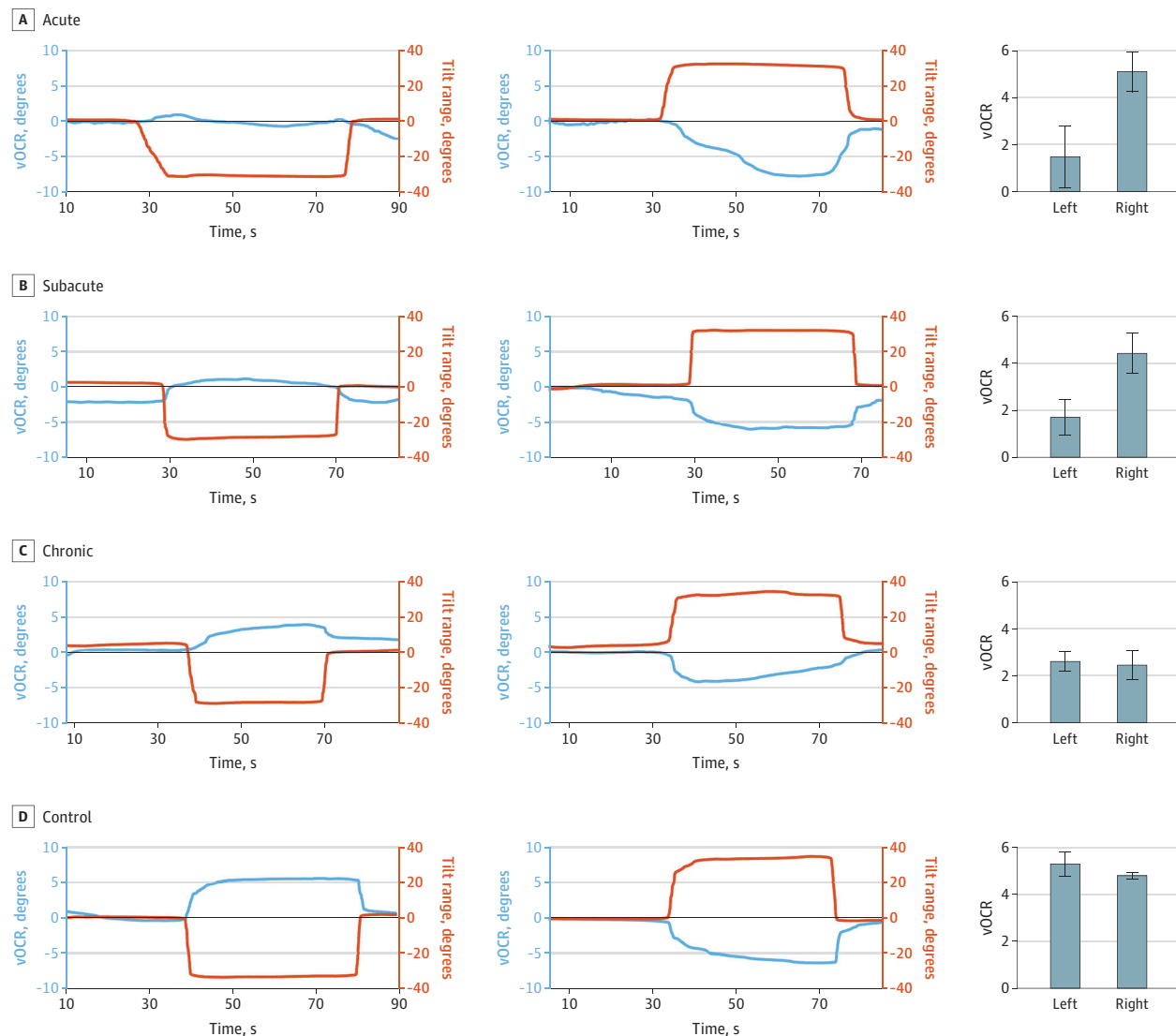
Figure 4 shows examples of vOCR measurement in patients with acute, subacute, and chronic vestibular loss as well as a healthy control individual. In the patient with acute loss, there was a vOCR asymmetry with 1.5° on the side of vestibular loss and 5° on the healthy side. In the patient with subacute loss, there was also an asymmetry, with a vOCR of 1.7° on the side of vestibular loss and 4.4° on the healthy side. In the patient with chronic loss, there was no significant asymmetry and vOCR was reduced at approximately 2.5° on both sides. In the healthy control individual, vOCR was approximately 5° on both sides. Thus, with 30° tilt, a vOCR value less than 4.5° indicated vestibular loss with more reduced values in patients with acute vestibular loss.

Discussion

In this study we investigated whether the vOCR can be used as a clinical test to detect loss of otolith function. Our findings show that the vOCR test can be easily applied with a bedside tilt of the head and torso en bloc. Using a 30° tilt, vOCR values less than 4.5° could detect patients with loss of vestibular function compared with healthy controls with an accuracy of 83%. In patients with acute loss, vOCR was lower on the side of vestibular loss, while the asymmetry was reduced in subacute and chronic groups. These findings suggest that the vOCR deficit is pronounced early following vestibular loss but can recover over time.

Under normal circumstances, when the head is stationary, the utricular inputs are in a state of equilibrium, similar to the balance between neural inputs from the semicircular canals at rest.^{2,26} Removal of the utricular inputs from one side can cause imbalance, leading to a pathologic OCR response, skew deviation of the eyes, and a roll head tilt toward the side

Figure 4. Examples of Video-Oculography Measurement of Ocular Counter-Roll (vOCR) in Individual Patients With Acute, Subacute, and Chronic Vestibular Loss as Well as a Healthy Control Individual



The vOCR recording is shown as the blue trace/axis and the tilt angle is shown as the orange trace/axis. The negative values on the graphs indicate leftward direction and the positive values indicate rightward direction. The mean vOCR values from 3 measurements on each side are shown in the bar graphs. The lower vOCR values correspond to the side of the vestibular loss with an overall lower asymmetry in the patient with chronic vestibular loss. A, In the patient with acute vestibular loss, there is an asymmetry with a vOCR of 1.5° on the side

of vestibular loss and 5° on the healthy side. B, In the patient with subacute vestibular loss, there is also an asymmetry with a vOCR of 1.7° on the side of vestibular loss and 4.4° on the healthy side. C, In the patient with chronic right-side vestibular loss, there is no significant asymmetry and vOCR values are reduced by approximately 2.4° and 2.6°. D, In the healthy control individuals, vOCR values are 5.3° and 4.7°.

of vestibular loss.^{12,27-31} Such otolith-related findings are compensated as presumably the balance in neural activity between the 2 vestibular nuclei is restored to a new set point over time, just as the spontaneous nystagmus from canal imbalance also declines following unilateral vestibular loss.^{11,32,33} Consistent with such recovery, in this study we found symmetric reduction in vOCR on both sides in patients with chronic vestibular loss, but the vOCR deficits in those with acute vestibular loss were asymmetric with more reduced values on the side of vestibular loss. Future studies should investigate whether such recovery in vOCR corresponds with improve-

ment of daily function and vestibular symptoms in a same group of patients. If a significant association is found, vOCR can be used to track recovery in addition to detecting loss of otolith function. The noninvasive and quick application of the vOCR test can facilitate follow-up examinations for tracking recovery in patients.

The differences in vOCR among patients and healthy controls show that the vOCR can be valuable as a clinical test of otolith function to detect the extent and pattern of otolith hypofunction. Otolith pathways are commonly affected in vestibular disorders and, although vHIT has been widely used for

clinical assessment of canal function, a VOG method for evaluation of otolith-ocular function has been lacking. Similar to vHIT, vOCR is measured using high-speed VOG goggles and automated analysis software providing immediate results. Thus, if combined, these tools can be used as one VOG battery for evaluation of both otolith and canal functions. Such a VOG battery can also help to detect the side of otolith hypofunction as the vOCR on the nonlesion side can be reduced especially in patients with chronic loss of vestibular function (Figure 3 and Figure 4). In such cases with symmetric vOCR deficit, vHIT findings can be helpful to detect the side of vestibular hypofunction, whereas asymmetric vOCR deficits would be consistent with unilateral loss of otolith-ocular function with a lower vOCR value on the side of vestibular loss (Figure 4). Based on our results, the vOCR threshold to detect loss of otolith function was 4.5°. Future studies should examine patients with other causes and varying degrees of peripheral vestibular loss to establish more accurate thresholds for detecting loss of otolith-ocular function.

In this study we did not include dynamic vOCR during lateral movement of the head because it would result in stimulation of both the canals and otoliths, and thus the outcome would not be specific to the otoliths. A previous study compared vOCR during a static head position with the laboratory-based, vestibular-evoked myogenic potentials measurements of both utricular and saccular functions.²¹ Among various measurements, the click and tap ocular VEMP correlated with the vOCR values, consistent with all being measures of otolith-ocular or utricular function.²¹ Various methods have been reported for OCR measurement, using either VOG or scleral search coils, but the real-time vOCR measurement with a bedside maneuver was not used as a clinical test of otolith function.^{3,15,21,33-36} Another method for assessing otolith function is through the VOR generated by translation of the head.³⁷⁻³⁹ The normal VOR response during translation of the head is usually less than compensatory and it is often supplemented by saccade, which depending on the viewing

distance, can facilitate bringing the eyes to the target.⁴ Also, the manual head translation for VOR measurement can be contaminated by unintended head rotation that stimulates the semicircular canals.³⁹ Thus, using the translational VOR as a bedside test of otolith function is not as feasible as applying static head tilt for vOCR measurement.

Limitations

The present study has limitations. The sample size was small and the case-control design can overestimate the measures of test performance. In addition, vOCR was recorded at different time points in separate groups of patients and there was no direct measure of compensation or the outcome of interventions such as vestibular physical therapy to link with plausible vOCR recovery among patients. The heterogeneity of the vOCR deficit among the patient groups can also affect the accuracy of vOCR thresholds in the ROC analysis as the deficit becomes more symmetric in later stages of recovery. In addition, our sample was limited to patients with vestibular nerve resection and only 1 tilt angle for vOCR measurement (ie, 30°). Future studies should examine patients with larger tilt angles and include patients with other causes and varying degrees of peripheral vestibular loss to establish more accurate thresholds for detecting loss of otolith-ocular function.

Conclusions

In this study, our findings suggest that vOCR measurement can be easily applied as a bedside test with a simple tilt maneuver to detect or track loss of otolith-ocular function at different stages of compensation. Although patients with acute vestibular loss had lower vOCR values on the side of loss, patients with more chronic loss had symmetrically reduced vOCR values. Such a symmetrical vOCR deficit can be a result of long-term central adaptive processes and may also correspond with the state of recovery in patients.

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Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Sadeghpour, Otero-Millan, Kheradmand.

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REFERENCES

- MacDougall HG, Weber KP, McGarvie LA, Halmagyi GM, Curthoys IS. The video head impulse test: diagnostic accuracy in peripheral vestibulopathy. *Neurology*. 2009;73(14):1134-1141. doi:10.1212/WNL.0b013e3181bacf85
- Halmagyi GM, Chen L, MacDougall HG, Weber KP, McGarvie LA, Curthoys IS. The video head impulse test. *Front Neurol*. 2017;8:258. doi:10.3389/fneur.2017.00258
- Diamond SG, Markham CH. Ocular counterrolling as an indicator of vestibular otolith function. *Neurology*. 1983;33(11):1460-1469. doi:10.1212/WNL.33.11.1460
- Ramat S, Zee DS. Ocular motor responses to abrupt interaural head translation in normal humans. *J Neurophysiol*. 2003;90(2):887-902. doi:10.1152/jn.01121.2002

5. Bronstein AM, Gresty MA. Short latency compensatory eye movement responses to transient linear head acceleration: a specific function of the otolith-ocular reflex. *Exp Brain Res*. 1988;71(2):406-410. doi:10.1007/BF00247500
6. Basta D, Todt I, Scherer H, Clarke A, Ernst A. Postural control in otolith disorders. *Hum Mov Sci*. 2005;24(2):268-279. doi:10.1016/j.humov.2005.04.002
7. Hafstrom A, Fransson P-A, Karlberg M, Magnusson M. Subjective visual tilt and lateral instability after vestibular deafferentation. *Acta Otolaryngol*. 2006;126(11):1176-1181. doi:10.1080/00016480600684423
8. Smith PF. The growing evidence for the importance of the otoliths in spatial memory. *Front Neural Circuits*. 2019;13:66. doi:10.3389/fncir.2019.00066
9. Kheradmand A, Winnick A. Perception of upright: multisensory convergence and the role of temporo-parietal cortex. *Front Neurol*. 2017;8:552. doi:10.3389/fneur.2017.00552
10. Curthoys IS, Manzari L. Otolithic disease: clinical features and the role of vestibular evoked myogenic potentials. *Semin Neurol*. 2013;33(3):231-237. doi:10.1055/s-0033-1354595
11. Kim H-A, Hong J-H, Lee H, et al. Otolith dysfunction in vestibular neuritis: recovery pattern and a predictor of symptom recovery. *Neurology*. 2008;70(6):449-453. doi:10.1212/01.wnl.0000297554.21221.a0
12. Halmagyi GM, Curthoys IS, Brandt T, Dieterich M. Ocular tilt reaction: clinical sign of vestibular lesion. *Acta Otolaryngol Suppl*. 1991;481:47-50. doi:10.3109/00016489109131342
13. Collewin H, Van der Steen J, Ferman L, Jansen TC. Human ocular counterroll: assessment of static and dynamic properties from electromagnetic scleral coil recordings. *Exp Brain Res*. 1985;59(1):185-196. doi:10.1007/BF00237678
14. Otero-Millan J, Kheradmand A. Upright perception and ocular torsion change independently during head tilt. *Front Hum Neurosci*. 2016;10:573. doi:10.3389/fnhum.2016.00573
15. Otero-Millan J, Roberts DC, Lasker A, Zee DS, Kheradmand A. Knowing what the brain is seeing in three dimensions: a novel, noninvasive, sensitive, accurate, and low-noise technique for measuring ocular torsion. *J Vis*. 2015;15(14):11. doi:10.1167/15.14.11
16. Markham CH, Diamond SG. Ocular counterrolling in response to static and dynamic tilting: implications for human otolith function. *J Vestib Res*. 2002-2003;12(2-3):127-134.
17. Schmid-Priscoveanu A, Straumann D, Kori AA. Torsional vestibulo-ocular reflex during whole-body oscillation in the upright and the supine position; I: responses in healthy human subjects. *Exp Brain Res*. 2000;134(2):212-219. doi:10.1007/s002210000436
18. Curthoys IS. A critical review of the neurophysiological evidence underlying clinical vestibular testing using sound, vibration and galvanic stimuli. *Clin Neurophysiol*. 2010;121(2):132-144. doi:10.1016/j.clinph.2009.09.027
19. Pansell T, Ygge J, Schworm HD. Conjugacy of torsional eye movements in response to a head tilt paradigm. *Invest Ophthalmol Vis Sci*. 2003;44(6):2557-2564. doi:10.1167/iiov.02-0987
20. Uchino Y, Kushi K. Differences between otolith- and semicircular canal-activated neural circuitry in the vestibular system. *Neurosci Res*. 2011;71(4):315-327. doi:10.1016/j.neures.2011.09.001
21. Otero-Millan J, Treviño C, Winnick A, Zee DS, Carey JP, Kheradmand A. The video ocular counter-roll (vOCR): a clinical test to detect loss of otolith-ocular function. *Acta Otolaryngol*. 2017;137(6):593-597. doi:10.1080/00016489.2016.1269364
22. Strupp M, Arbusow V, Dieterich M, Sautier W, Brandt T. Perceptual and oculomotor effects of neck muscle vibration in vestibular neuritis: ipsilateral somatosensory substitution of vestibular function. *Brain*. 1998;121(pt 4):677-685. doi:10.1093/brain/121.4.677
23. Karlberg M, Aw ST, Black RA, Todd MJ, MacDougall HG, Halmagyi GM. Vibration-induced ocular torsion and nystagmus after unilateral vestibular deafferentation. *Brain*. 2003;126(pt 4):956-964. doi:10.1093/brain/awg091
24. McKenna GJ, Peng GYC, Zee DS. Neck muscle vibration alters visually perceived roll in normals. *J Assoc Res Otolaryngol*. 2004;5(1):25-31. doi:10.1007/s10162-003-4005-2
25. Bronstein AM, Morland AB, Ruddock KH, Gresty MA. Recovery from bilateral vestibular failure: implications for visual and cervico-ocular function. *Acta Otolaryngol Suppl*. 1995;520(Pt 2):405-407. doi:10.3109/00016489509125283
26. Chang T-P, Zee DS, Kheradmand A. Technological advances in testing the dizzy patient: the bedside examination is still the key to successful diagnosis. In: Gleason AT, Kesser BW, eds. *Dizziness and Vertigo Across the Lifespan*. Elsevier; 2019:9-30. doi:10.1016/B978-0-323-55136-6.00002-2
27. Brodsky MC, Donahue SP, Vaphiades M, Brandt T. Skew deviation revisited. *Surv Ophthalmol*. 2006;51(2):105-128. doi:10.1016/j.survophthal.2005.12.008
28. Kheradmand A, Colpak AI, Zee DS. Eye movements in vestibular disorders. *Handb Clin Neurol*. 2016;137:103-117. doi:10.1016/B978-0-444-63437-5.00008-X
29. Kheradmand A, Zee DS. The bedside examination of the vestibulo-ocular reflex (VOR): an update. *Rev Neurol (Paris)*. 2012;168(10):710-719. doi:10.1016/j.neurol.2012.07.011
30. Cherchi M. Utricular function in vestibular neuritis: a pilot study of concordance/discordance between ocular vestibular evoked myogenic potentials and ocular cyclopolygon. *Exp Brain Res*. 2019;237(6):1531-1538. doi:10.1007/s00221-019-05529-8
31. Halmagyi GM, Gresty MA, Gibson WP. Ocular tilt reaction with peripheral vestibular lesion. *Ann Neurol*. 1979;6(1):80-83. doi:10.1002/ana.410060122
32. Goto F, Ban Y, Tsutsumi T. Acute audiovestibular deficit with complete ocular tilt reaction and absent VEMPs. *Eur Arch Otorhinolaryngol*. 2011;268(7):1093-1096. doi:10.1007/s00405-011-1574-2
33. Schmid-Priscoveanu A, Straumann D, Böhmer A, Obzina H. Vestibulo-ocular responses during static head roll and three-dimensional head impulses after vestibular neuritis. *Acta Otolaryngol*. 1999;119(7):750-757. doi:10.1080/00016489950180379
34. Kingma H, Stegeman P, Vogels R. Ocular torsion induced by static and dynamic visual stimulation and static whole body roll. *Eur Arch Otorhinolaryngol*. 1997;254(suppl 1):S61-S63. doi:10.1007/BF02439726
35. Schworm HD, Ygge J, Pansell T, Lennerstrand G. Assessment of ocular counterroll during head tilt using binocular video oculography. *Invest Ophthalmol Vis Sci*. 2002;43(3):662-667.
36. Zingler VC, Kryvoshey D, Schneider E, Glasauer S, Brandt T, Strupp M. A clinical test of otolith function: static ocular counterroll with passive head tilt. *Neuroreport*. 2006;17(6):611-615. doi:10.1097/00001756-200604240-00011
37. Lempert T, Gianna C, Brookes G, Bronstein A, Gresty M. Horizontal otolith-ocular responses in humans after unilateral vestibular deafferentation. *Exp Brain Res*. 1998;118(4):533-540. doi:10.1007/s002210050309
38. Kessler P, Tomlinson D, Blakeman A, Rutka J, Ranalli P, Wong A. The high-frequency/acceleration head heave test in detecting otolith diseases. *Otol Neurotol*. 2007;28(7):896-904. doi:10.1097/MAO.0b013e3181256543
39. Ramat S, Zee DS, Minor LB. Translational vestibulo-ocular reflex evoked by a "head heave" stimulus. *Ann N Y Acad Sci*. 2001;942:95-113. doi:10.1111/j.1749-6632.2001.tb03738.x